

Juan Segundo Schmid Aguerre

Python Backend & AI Engineer

Buenos Aires, Argentina · Remote · GMT-3, full overlap with US Eastern business hours · English: professional working proficiency · Spanish: native

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SUMMARY

Backend engineer building **production systems in Python and FastAPI**, focused on hard third-party integrations, multi-tenant architecture and **LLM / Document-AI pipelines**. I designed, built and operate **Spom Bridge**, a multi-tenant e-invoicing SaaS live in production integrating Argentina's tax authority (AFIP/ARCA SOAP), Mercado Libre, Tiendanube and Mercado Pago — architecture, backend, front-end, auth, observability and deploy, on my own. At Scienza I build Python and Document-AI automation for pharma operations, cutting operational time by up to 70%. Seeking remote backend / AI roles where the hard part is integration, resilience and data.

EXPERIENCE

Founder & Lead Engineer 2025 – Present

Spom Bridge (Spom Solutions) · Independent SaaS product, live in production

- Architected and operate a **multi-tenant SaaS in production** on FastAPI, PostgreSQL and Redis, containerized with Docker on Railway — 40+ domain modules with strict per-tenant data isolation. Also built the **React 18 + TypeScript** front-end end to end (Vite, Tailwind CSS).
- Integrated the **AFIP/ARCA SOAP web services** (WSAA certificate signing, WSFE invoice authorization) so every sale is invoiced with no manual step. The tax API is ambiguous and non-idempotent by design — solved with per-document idempotency keys, token caching, bounded retries and a reconciliation state machine.
- Built **OAuth2 connectors and signed-webhook consumers** for Mercado Libre, Tiendanube and Mercado Pago, keeping orders, catalog and payments in sync in near real time.
- Shipped production-grade security (JWT, 2FA, CSRF protection, rate limiting) and held quality with **250+ automated tests**, CI gating every merge and Sentry-based observability.

Automation & AI Developer Feb 2025 – Present

Scienza Argentina · Pharmaceutical distribution

- Re-architected the pharmacy **document-reception pipeline with Document AI and OCR**, removing manual data entry from a high-volume, compliance-sensitive process.
- Layered an **LLM over raw OCR output** to raise extraction accuracy and emit clean, structured JSON consumed directly by downstream enterprise systems.
- Built Python automations that **cut operational time by up to 70%** across business-critical workflows, integrating SAP and exposing Flask REST APIs between enterprise systems.
- Delivered Pandas/SQL pipelines feeding automated executive dashboards, with measurable ROI presented to **C-level stakeholders**.

Functional Analyst Jan 2024 – Feb 2025

Whirlpool Corporation · Fortune 500 manufacturing

- Gathered and documented requirements for digital transformation projects, translating business needs into technical specifications for development teams.
- Designed system and process improvements under **Agile/Scrum**, coordinating IT and business stakeholders from analysis through go-live.

SELECTED PROJECTS

- OCR + LLM Document Processing** — Ingests documents, runs AWS Textract OCR and layers an LLM to raise extraction accuracy, returning clean structured JSON. *Python, Flask, Next.js, AWS Textract.*
- Automated WordPress Publisher** — Python pipeline that generates SEO content with an LLM (Groq) and publishes via the WordPress REST API. *Python, LLM, REST API.*
- Inventory Management System** — Full-stack platform with role-based access control, CRUD and reporting. *Java, Spring Boot, SQL, React.*

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Java
Backend	FastAPI, Flask, Django, SQLAlchemy (async), REST API design, OAuth2, JWT/2FA, webhooks, SOAP integrations, multi-tenant architecture
AI & LLM	Claude API (agents, tool use), LLM-assisted extraction and structuring, Google Document AI, AWS Textract, prompt engineering, Claude Code
Data	PostgreSQL, Redis, MongoDB, SQL Server, Pandas, NumPy
Front-end	React 18, TypeScript, Vite, Tailwind CSS, Next.js
Infra & QA	Docker, Railway, CI/CD, Sentry, Git, Pytest, Vitest, Ruff, ESLint

EDUCATION

Systems Engineering (in progress) & **University Technician in Software Development** (completed) — UADE, Buenos Aires 2021 – Present